FIN301 | FINANCIAL ANALYTICS

Short Notes: Model Answers

25 questions • 3 marks each • about 100 words per answer

> Each answer gives the definition, the formula where one exists, a Financial Analytics or stock-market example, and its interpretation or business use, as the question paper asks. Example figures are illustrative.

Section A — Statistical Analysis in R

1. Data Summary: Mean and Median

Q. *Differentiate mean and median. If a company's sales data are strongly right-skewed, which measure may better represent a typical company and why?*

The **mean** is the sum of all values divided by their number, so every value, including extreme ones, affects it. The **median** is the middle value of the sorted data and is not pulled by extremes.

In right-skewed sales data a few very large companies pull the mean upward, so the **median** better represents a typical company. **Example:** sales (₹ crore) of 10 firms are 10, 15, 20, 25, 30, 35, 40, 45, 50 and 2,000. Mean = 227, but median = (30 + 35) ÷ 2 = 32.5, which is far closer to what most firms actually sell.

2. Variance and Standard Deviation

Q. *Explain the relationship between variance and standard deviation. How can standard deviation be used to compare the volatility of two stocks measured over the same period?*

Variance is the average squared deviation of values from their mean: $s^2 = \Sigma(x - \bar{x})^2 \div (n - 1)$. **Standard deviation (SD)** is its square root, SD = √variance, which brings the measure back to the original unit (e.g. % return), making it easier to interpret.

SD of returns measures **volatility (total risk)**. **Example:** over the same year, Stock A's daily returns have SD 1.2% and Stock B's 2.5%; B's returns swing about twice as widely, so B is more volatile and riskier. The comparison is valid only if both use the same period and return frequency.

3. Coefficient of Variation

Q. *Explain the meaning and importance of the coefficient of variation (CV) when comparing variability in the returns of two stocks.*

CV = Standard deviation ÷ Mean × 100. It measures **relative** variability, i.e. risk per unit of average return, and is unit-free.

It matters when two stocks have different average returns, because a larger SD alone does not show which is riskier relative to its reward. **Example:** Stock A: mean 12%, SD 6% → CV = 50%. Stock B: mean 20%, SD 15% → CV = 75%. B has higher return but takes more risk per unit of return, so A is more efficient. **Caution:** CV is misleading when the mean is close to zero or negative.

4. Interquartile Range (IQR)

Q. What does the interquartile range measure? Explain how it can help identify unusually high or low observations in a financial variable such as sales or loan amount.

IQR = Q3 − Q1, the spread of the middle 50% of the data. Because it ignores the extreme quarters, it is not affected by outliers.

Outliers are identified with fences: **lower fence = Q1 − 1.5 × IQR** and **upper fence = Q3 + 1.5 × IQR**; values outside are unusual. **Example:** loan amounts have Q1 = ₹2 lakh and Q3 = ₹6 lakh, so IQR = ₹4 lakh and the upper fence = 6 + 1.5 × 4 = ₹12 lakh. A loan of ₹25 lakh is flagged and checked for a data-entry error or a large-ticket risk exposure.

5. Skewness and Kurtosis

Q. What do skewness and kurtosis tell an analyst about a distribution? Explain why these measures may be useful when analysing loan amounts or stock returns.

Skewness measures asymmetry. Positive skew = long right tail (mean > median); negative skew = long left tail. **Kurtosis** measures tail heaviness: normal = 3; above 3 = **fat tails**, i.e. extreme values occur more often than the normal curve predicts.

Use: loan amounts are usually positively skewed (many small loans, a few very large ones), so the median and log transformations suit them better than the mean. Stock returns often show negative skew and high kurtosis, meaning sudden large losses are more likely than SD suggests, so risk models (e.g. VaR) that assume normality may understate risk.

6. Covariance Matrix

Q. What is a covariance matrix? Explain its importance in financial analysis using a portfolio of two or more stocks as an example.

A **covariance matrix** is a square, symmetric table showing the covariance between every pair of variables; the **diagonal** contains each variable's variance. Positive covariance means two stocks tend to move together; negative means they move in opposite directions.

It is the key input for **portfolio risk**. For two stocks: $\sigma p^2 = wA^2\sigma A^2 + wB^2\sigma B^2 + 2wA \cdot wB \cdot Cov(A, B)$. If a bank stock and a pharma stock have low or negative covariance, combining them gives portfolio risk lower than the weighted average of their individual risks (**diversification**). It is used in Markowitz portfolio optimisation; in R: `cov(returns)`.

7. Correlation and Regression

Q. Differentiate correlation and regression. Give one financial example in which correlation measures association while regression is used to estimate or predict an outcome.

Correlation (r) measures the strength and direction of the linear association between two variables, from −1 to +1; it is symmetric and does not distinguish dependent and independent variables. **Regression** fits an equation (Y = a + bX) in which a dependent variable is explained by one or more independent variables, so it can **estimate and predict**.

Example: the correlation between a stock's returns and Nifty returns is 0.80 (strong positive association). Regressing stock return on Nifty return gives beta b = 1.2, predicting that the stock moves about 1.2% for every 1% move in the market. Neither proves causation.

8. Null and Alternative Hypotheses

Q. *Explain the null and alternative hypotheses in a statistical test. Illustrate them with a test of whether the mean returns of two investment groups differ.*

The **null hypothesis (H_o)** is the statement of no effect or no difference; it is assumed true unless the data provide strong evidence against it. The **alternative hypothesis (H_1)** is the claim the analyst wants evidence for.

Example: comparing large-cap funds (A) and mid-cap funds (B): **H_o: $\mu A = \mu B$** (mean returns are equal); **H_1: $\mu A \neq \mu B$** (they differ – two-tailed). If the test gives $p < 0.05$, H_o is rejected and the mean returns are concluded to differ; otherwise we **fail to reject** H_o (we do not 'prove' it true).

9. Independent-Sample t-Test

Q. *What is an independent-sample t-test? Explain how it could be used to compare the average returns of two independent investment groups or the average loan amounts of two customer groups.*

An **independent-sample t-test** checks whether the means of two **unrelated** groups differ significantly. $t = (\bar{x}_1 - \bar{x}_2) \div$ standard error of the difference. It assumes independent observations and roughly normal data (Welch's version does not need equal variances).

Example: a bank compares the average loan amount of salaried customers (₹8.2 lakh) and self-employed customers (₹9.6 lakh). H_o: the means are equal. In R: `t.test(amount ~ group, data = loans)`. If $p < 0.05$, the difference is statistically significant and may justify separate credit policies; if not, the gap may be due to chance. Fund returns of two groups are compared the same way.

10. P-Value and Significance Level

Q. *What is a p-value? Explain how it is used with a 5% significance level to make a statistical decision, and state one common misinterpretation to avoid.*

The **p-value** is the probability of obtaining a result at least as extreme as the one observed, **assuming H_o is true**. A small p-value means the data are unlikely under H_o.

Decision rule at $\alpha = 5\%$: if $p \leq 0.05$, reject H_o (statistically significant); if $p > 0.05$, fail to reject H_o. **Example:** testing whether two funds' mean returns differ gives $p = 0.02 < 0.05$, so H_o is rejected. **Misinterpretation to avoid:** the p-value is **not** the probability that H_o is true, and a small p-value does not mean the difference is large or economically important.

11. Type I and Type II Errors

Q. *Explain Type I and Type II errors using a Financial Analytics example, such as incorrectly flagging a good borrower as risky or failing to identify a risky borrower.*

A **Type I error** (false positive) is rejecting a true H_o; its probability is α (significance level). A **Type II error** (false negative) is failing to reject a false H_o; its probability is β, and power = $1 - \beta$.

Credit example (H_o: the borrower is good): Type I – a good borrower is wrongly flagged as risky and the loan is rejected, so the bank loses interest income and a customer. **Type II** – a risky borrower is classified as safe and approved, leading to default losses. The two trade off: a stricter model reduces Type II errors but increases Type I errors, so banks set the cut-off by comparing their costs.

12. Multiple Regression: Adjusted R^2

Q. *What does Adjusted R^2 indicate in a multiple regression model, and why can it be preferred to R^2 when comparing models with different numbers of predictors?*

R^2 shows the proportion of variation in the dependent variable explained by the predictors. **Adjusted R^2 = $1 - (1 - R^2)(n - 1) \div (n - k - 1)$** corrects it for the number of predictors (k) and sample size (n).

R^2 never falls when a predictor is added, even a useless one, so it favours bigger models. Adjusted R^2 rises only if a new variable improves the fit more than would be expected by chance. **Example (n = 30):** adding a fifth ratio to a Z-score model raises R^2 from 0.800 to 0.805, but adjusted R^2 falls from 0.768 to 0.764, so the simpler four-variable model is preferred.

13. Multicollinearity and VIF

Q. *What is multicollinearity in multiple regression? Explain how a high Variance Inflation Factor (VIF) can affect coefficient estimates and their interpretation.*

Multicollinearity occurs when two or more independent variables are highly correlated with each other, so they carry overlapping information. **VIF = $1 \div (1 - R_j^2)$**, where R_j^2 comes from regressing that predictor on the other predictors; VIF above 5 (or 10) signals a problem.

A high VIF **inflates standard errors** (VIF = 10 makes the SE about $\sqrt{10} \approx 3.2$ times larger), so coefficients become unstable, may show wrong signs and appear insignificant even when R^2 is high, making it hard to judge each variable's separate effect. **Example:** EBIT/TA and retained earnings/TA in a Z-score model. Remedies: drop or combine variables, or collect more data.

Section B — Technical Analysis

14. Price Charts and Trend Context

Q. *Why should a technical analyst examine a stock's price chart and prevailing trend before interpreting individual technical indicators?*

Indicators are calculated from price, so they summarise it but cannot replace it. The price chart shows the **trend (up, down or sideways), support and resistance, volatility and key events**, which decide how an indicator should be read.

The same reading means different things in different trends. **Example:** RSI above 70 in a strong uptrend often reflects strong momentum and can persist, whereas in a sideways market it may signal a reversal; an 'oversold' RSI in a strong downtrend may keep falling. Oscillators work best in ranges and moving-average signals in trends, so checking the trend first avoids false signals and keeps trades in the direction of the trend.

15. Simple, Exponential and Weighted Moving Averages

Q. *Differentiate Simple Moving Average (SMA), Exponential Moving Average (EMA) and Weighted Moving Average (WMA) in terms of how they weight historical prices.*

SMA: gives **equal weight** to each of the last n prices; smooth but slowest to react. **WMA:** uses **linearly declining weights** (n for the latest price, n − 1 for the previous, … 1 for the oldest) over the last n prices. **EMA:** gives **exponentially declining weights** to all past prices, $EMA_t = \alpha \cdot P_t + (1 - \alpha) \cdot EMA_{t-1}$ with $\alpha = 2 \div (n + 1)$; it reacts fastest to new prices.

Example: 3-day prices 100, 102, 106 (oldest to latest): SMA = 308 ÷ 3 = 102.67; WMA = (1×100 + 2×102 + 3×106) ÷ 6 = 103.67, closer to the latest price. EMA and WMA follow trends faster but give more false signals.

16. Price Relative to a Moving Average

Q. *What can the relationship between a stock's closing price and its 200-day moving average indicate about the long-term trend? Mention one limitation of this signal.*

The **200-day moving average (200-DMA)** represents the long-term average price. A close **above** a rising 200-DMA indicates a **long-term uptrend** (bullish); a close **below** a falling 200-DMA indicates a **long-term downtrend** (bearish). A cross of the price through the 200-DMA may mark a change of trend, and the distance from it shows how stretched the price is. Many institutional investors use it as a trend filter.

Limitation: it is a **lagging** indicator, built from 200 past prices, so it confirms a trend only after much of the move has happened; in sideways markets the price repeatedly crosses it, producing false (whipsaw) signals.

17. Golden Cross

Q. *What is a Golden Cross and what information does it provide to a technical analyst?*

A **Golden Cross** occurs when a **short-term moving average crosses above a long-term moving average**, classically the **50-day MA crossing above the 200-day MA**.

It indicates that recent prices are now higher than the long-term average, i.e. momentum has shifted from bearish to **bullish**, and it may mark the start of a long-term uptrend. Analysts use it as a **buy or confirmation signal**, especially when supported by rising volume and a rising 200-day MA. **Caution:** because both averages lag, the cross appears after part of the rally has already occurred, and in sideways markets it can be a false signal followed quickly by a Death Cross.

18. Bearish Cross / Death Cross

Q. *What is a Bearish Cross (Death Cross), and what information does it provide to a technical analyst?*

A **Death Cross** occurs when a **short-term moving average crosses below a long-term moving average**, classically the **50-day MA falling below the 200-day MA**.

It indicates that recent prices are below the long-term average, so momentum has turned **bearish** and a long-term downtrend may be starting. Analysts use it as a **sell, exit or caution signal** – reducing long positions, tightening stop-losses or avoiding new purchases – particularly when confirmed by high volume and a falling 200-day MA. **Caution:** it lags price and sometimes appears near the end of a decline, and in range-bound markets it is often reversed by a Golden Cross (whipsaw).

19. Relative Strength Index (RSI)

Q. *Explain RSI and how it can be used to develop a trading strategy.*

RSI (Welles Wilder) is a momentum oscillator from 0 to 100 that compares the size of recent gains with recent losses, usually over **14 periods**: **RSI = 100 – 100 ÷ (1 + RS)**, where RS = average gain ÷ average loss. In R: `TTR::RSI(close, n = 14)`.

Strategy: buy when RSI rises back **above 30** from oversold and sell or exit when it falls back **below 70** from overbought; in an uptrend, buy pullbacks where RSI holds near 40–50. Rules are improved with a **trend filter** (e.g. take long trades only above the 200-DMA), divergence signals and a stop-loss, and should be back-tested before use.

20. RSI Overbought and Oversold Levels

Q. *Explain the conventional RSI thresholds of 70 and 30. Why should an analyst avoid treating an overbought reading as an automatic sell signal or an oversold reading as an automatic buy signal?*

RSI **above 70** is conventionally **overbought** (prices have risen sharply and may be stretched); RSI **below 30** is **oversold** (prices have fallen sharply and may rebound).

These are warnings, not signals. In a **strong uptrend** RSI can stay above 70 for weeks while the price keeps rising, so selling at the first reading means missing the trend; in a **strong downtrend** RSI can stay below 30 as prices keep falling. Analysts therefore wait for **confirmation** – RSI crossing back below 70 or above 30, a price break, or divergence – and consider the trend; some use 80/20 in strongly trending markets.

21. RSI Divergence

Q. *What is bullish or bearish divergence between price and RSI? Explain what it may indicate and why confirmation is still important.*

Divergence occurs when price and RSI move in opposite directions. **Bearish divergence:** price makes a **higher high** but RSI makes a **lower high**, showing weakening upward momentum and a possible reversal down. **Bullish divergence:** price makes a **lower low** but RSI makes a **higher low**, showing that selling pressure is fading and a possible reversal up.

Confirmation is important because divergence shows only that momentum is slowing, not that the trend has ended; in strong trends it can persist over several swings and give early or false signals. Traders wait for a price break of a trendline or support/resistance, RSI crossing 50, or a moving-average crossover, and use a stop-loss.

22. Moving Average Convergence Divergence (MACD)

Q. *Explain MACD and how it can be used in making trading decisions.*

MACD (Gerald Appel) is a trend-following momentum indicator: **MACD line = 12-period EMA − 26-period EMA**; **signal line = 9-period EMA of the MACD line**; **histogram = MACD − signal**. A positive MACD means short-term prices are above the longer-term average (bullish momentum).

Trading uses: (1) **signal-line crossovers** – MACD crossing above the signal line suggests buying, below suggests selling; (2) **zero-line crossovers** confirm a change of trend; (3) **divergence** between MACD and price warns of a reversal; (4) a shrinking histogram shows momentum fading. In R: `TTR::MACD(close, 12, 26, 9)`. It lags and gives false signals in sideways markets.

23. MACD Crossover

Q. *Explain the difference between a bullish and a bearish MACD crossover. How might a trader use the crossover alongside the prevailing price trend?*

A **bullish crossover** occurs when the **MACD line crosses above the signal line** (histogram turns positive), showing rising upward momentum. A **bearish crossover** occurs when the **MACD line crosses below the signal line** (histogram turns negative), showing rising downward momentum.

Crossovers are most reliable **in the direction of the main trend**. In an uptrend (e.g. price above its 200-DMA), a trader takes bullish crossovers as entry signals – especially after a pullback – and uses bearish crossovers only to tighten stops; in a downtrend the opposite applies. A crossover that also occurs on the correct side of the zero line, with volume support, gives stronger confirmation.

24. Average True Range (ATR)

Q. *Explain how ATR is used for setting a stop-loss level in a trading strategy.*

ATR (Wilder) measures volatility as the average **true range** over usually 14 periods, where true range = the largest of (High − Low), |High − previous close| and |Low − previous close|. It is expressed in rupees.

For a stop-loss, the trader places the stop a multiple of ATR away from the entry: **long stop = entry price − k × ATR** (k is typically 1.5–3). **Example:** buy at ₹1,500 with ATR = ₹30 and k = 2: stop-loss = 1,500 − 60 = **₹1,440**. This gives volatile stocks wider stops and calm stocks tighter stops, avoiding exits caused by normal noise; it can also be trailed (highest close − k × ATR) and used to size positions.

25. Bollinger Bands and Signal Confirmation

Q. *What are Bollinger Bands, and how can an analyst use them to assess relative price position and changing volatility? State one caution when interpreting a band touch. You may also explain why moving-average crossovers can produce lagging or false signals and how confirmation can help.*

Bollinger Bands (John Bollinger) consist of a **20-day SMA** (middle band) and **upper/lower bands at ± 2 standard deviations**. **Relative position:** price near the upper band is high relative to recent prices, near the lower band is low (%B measures this). **Volatility:** bands **widen** when volatility rises and **narrow** (a 'squeeze') when it falls, which often precedes a breakout.

Caution: a touch of the upper band is not automatically a sell signal; in strong trends prices can 'walk the band'. Similarly, moving-average crossovers lag because they use past prices and whipsaw in sideways markets, so signals should be confirmed with volume, RSI/MACD or a price breakout.